A university campus spread across **168 acres** with **50 buildings** and approximately **10,000 simultaneous users** requires a highly efficient and scalable communication network. The network must provide uninterrupted internet connectivity with minimal delay and high reliability.

To achieve this, a hybrid communication system combining **optical fiber backbone (wired communication)** and **Wi-Fi 6 wireless network** is proposed. Proper bandwidth estimation, access point planning, and topology selection are essential for optimal performance.

- Campus Area = 168 acres
- Number of Buildings = 50
- Total Users (Simultaneous) = 10,000
- Transmission Media = Wired + Wireless

To design the network, the following engineering assumptions are made:

- Average bandwidth per user = 5 Mbps (For video lectures, cloud access, browsing, online exams)
- Each Wi-Fi 6 access point supports ≈ 250 users
- Backbone link capacity between buildings = 10 Gbps (minimum)

Step 1: Total Required Bandwidth

$$\begin{aligned} &Total\ Bandwidth = Number\ of\ Users \times Bandwidth\ per\ User \\ &= 10{,}000 \times 5Mbps \\ &= 50{,}000Mbps \\ &= 50Gbps \end{aligned}$$

Step 2: Adding Future Growth Margin (30%)

$$50Gbps + 30\% = 65Gbps$$

Therefore, the required backbone internet capacity is:

$$\boxed{Minimum\ 60\text{–}70\ Gbps}$$

For future scalability, provisioning up to **100 Gbps optical fiber backbone** is recommended.

Since the campus area is large (168 acres), long-distance communication between buildings requires:

Transmission Medium:

- **Single-mode Optical Fiber**

Topology:

- **Ring Topology**

Reason for Ring Topology:

- Provides redundancy
- If one link fails, data travels in the opposite direction
- Reduces single point of failure

A centralized **Data Center** will contain:

- Core routers
- Layer-3 switches
- Firewalls
- Network management system

All 50 buildings will be connected using underground fiber cables.

Technology Used:

- Wi-Fi 6 (IEEE 802.11ax)
- Dual Band (2.4 GHz + 5 GHz)

Access Point Calculation:

$$Number\ of\ APs = \frac{Total\ Users}{Users\ per\ AP}$$
$$= \frac{10{,}000}{250}$$
$$= 40\ High - density\ AP\ zones$$